INTERNATIONAL ISLAMIC UNIVERSITY MALAYSIA

KULLIYYAH OF ENGINEERING

MECHATRONICS SYSTEM INTEGRATION (MCTA 3203)

SEMESTER 2 2025/2026

WEEK 5: CONTROLLING A 12V DC LINEAR ACTUATOR USING L298P MOTOR DRIVER AND RELAY

SECTION 1

GROUP 11

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Date of Experiment: Wednesday, 1st April 2026

Date of Submission: Wednesday, 8th April 2026

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ABSTRACT

This experiment shows the design and implementation of a control system for a 12V DC linear actuator utilizing an L298P motor driver and a dual-relay configuration. The primary objective was to achieve bi-directional motion control and speed regulation while integrating safety feedback mechanisms. The experiment was divided into two phases: first, establishing the fundamental H-bridge logic to manage polarity and employing Pulse Width Modulation (PWM) for variable speed control; and second, integrating a limit switch sensor to automate the stopping sequence and provide visual status via an LED. Results demonstrated that the H-bridge configuration effectively reversed motor polarity for extension and retraction, while the PWM signal successfully modulated the actuator's velocity. Furthermore, the integration of sensor feedback ensured precise positioning and operational safety by halting movement upon physical contact. This study confirms the efficacy of combining solid-state drivers with electromechanical relays to manage high-current linear motion tasks in automated systems.

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INTRODUCTION

Linear actuators are fundamental components in modern robotics and industrial automation, providing precise linear motion by converting the rotational energy of a DC motor into translational force. Controlling these devices requires a robust interface capable of handling high current and voltage levels, typically 12V or higher, which standard microcontrollers cannot provide directly. The L298P motor driver serves as a specialized H-bridge integrated circuit that allows for high-power switching and direction control. By leveraging the principles of electromagnetic induction and mechanical leverage, these systems enable automated tasks ranging from simple gate opening to complex robotic limb positioning.

In the first phase of this experiment, the focus was on mastering the directional control and speed modulation of the actuator. The H-bridge logic was explored to understand how diagonal switching patterns alter the polarity across the motor terminals, thereby switching between clockwise and counter-clockwise rotation (extension and retraction). To enhance control beyond simple "on/off" states, Pulse Width Modulation (PWM) was introduced. By rapidly toggling the enable pin of the L298P driver, the average voltage delivered to the actuator could be varied, allowing for smooth acceleration and deceleration, which is critical for reducing mechanical stress on the actuator’s internal gears.

The second task expanded the system’s complexity by transforming it into a "closed-loop" feedback system through sensor integration. While manual control is effective for basic tasks, industrial applications require the system to "sense" its environment to prevent mechanical failure. This was achieved by incorporating a limit switch and a dual-relay setup. The control logic was programmed to monitor the sensor's state; upon triggering the limit switch, the microcontroller would interrupt the power cycle, activate an LED indicator, and execute a timed pause. This task highlights the importance of hardware-software integration, demonstrating how mechanical inputs can be used to override motor commands to ensure safe and repeatable automated cycles.

TASK 1

1.0 MATERIALS AND EQUIPMENTS

- Cytron CT Uno
- Four Push Button
- DC Motor
- Battery 9V
- Breadboard
- Jumper Wires

1.1 EXPERIMENTAL SETUP

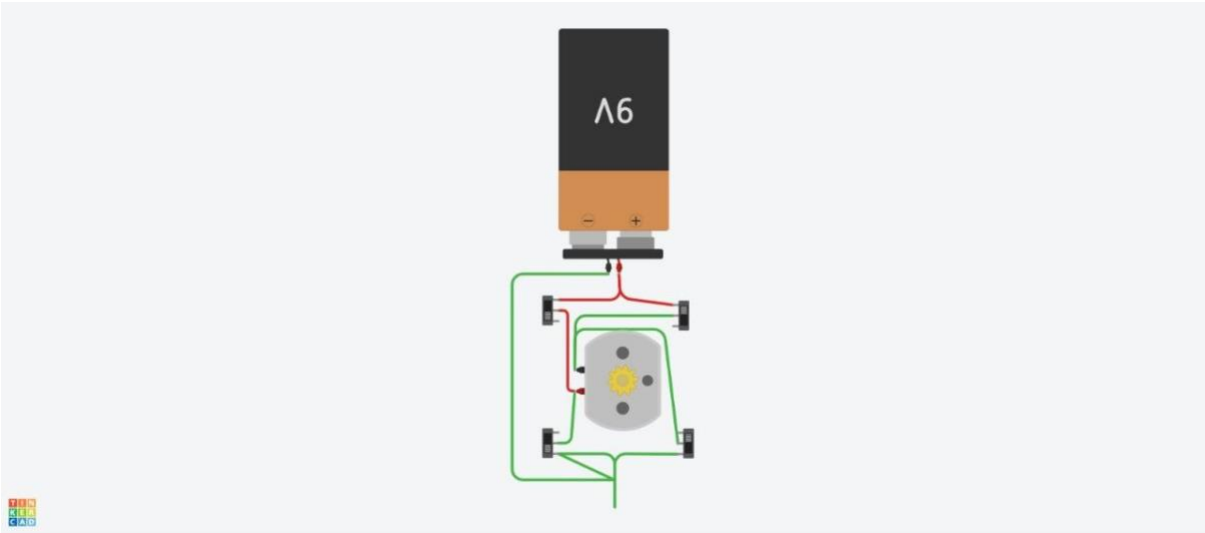

Fig.1 Connection of the DC Motor, Four Push Button and Battery

1. Place four push-button switches in an “H” pattern.
2. Connect the top two push button to the power source.
3. Connect the bottom two push button to the ground.
4. Place the DC motor in the centre of the “H”.
5. Connect the left terminal of the motor to the junction between top left and bottom left push button.
6. Connect the right terminal of the motor to the junction between top right and bottom right push button.

1.2 METHODOLOGY

a. Circuit assembly

- Connect the component as shown in Fig.1

b. Circuit logic

- The motor spins to the right direction when the top left button and the bottom right button pushed simultaneously.
- The motor spins to the left direction when the top right button and the bottom left button pushed simultaneously.

1.3 DATA COLLECTION

Push Button Condition	Motor Rotation
Top Left + Bottom Right	Clockwise
Top Right + Bottom Left	Anti-clockwise
Otherwise	Stopped

1.4 DATA ANALYSIS

The H-bridge was successfully converts a single polarity DC source into a bi-directional output through coordinated switching. When the top left button and the bottom right button are pressed, the polarity of the motor remain in one direction, resulting in clockwise rotation. When the top right button and the bottom left button are pressed, the polarity is reversed, causing the motor to rotate in anticlockwise direction.

1.5 RESULT

The experimental setup successfully demonstrated the fundamental principles of DC motor polarity reversal. By activating diagonal pairs of switches, the actuator able to perform two distinct mechanical action which is rotates to the right and to the left. When all buttons were in the OFF state, the actuator remained stationary.

1.6 DISCUSSION

The experiment validates that a four-switch H-bridge is a reliable method for controlling a linear actuator. The assembly correctly manages current flow to allow for bi-directional motion, provided that the diagonal switching logic is strictly maintained to avoid system shorts.

It is suggested to transform from a Tinkercad simulation to a physical circuit to ensure the system is efficient and achievable. After the basic circuit is successfully operated as required, replaced the push button with a dedicated H-bridge IC such as L298P and use a microcontroller to send a Pulse Width Modulation (PWM) signal to enable the actuator to move at varying speed.

TASK 2

Integrate any sensor into the circuit so that when it detects the actuator to a certain point it will activate an LED light and stop the linear actuator.

2.0 MATERIALS AND EQUIPMENTS.

- Cytron CT-UNO
- Two Way channel Relay
- DC 12V Linear Actuator
- 12V Power Supply
- Jumper wires
- Breadboard
- Switch
- Led light
- 220 Ω resistor

2.1 EXPERIMENTAL SETUP

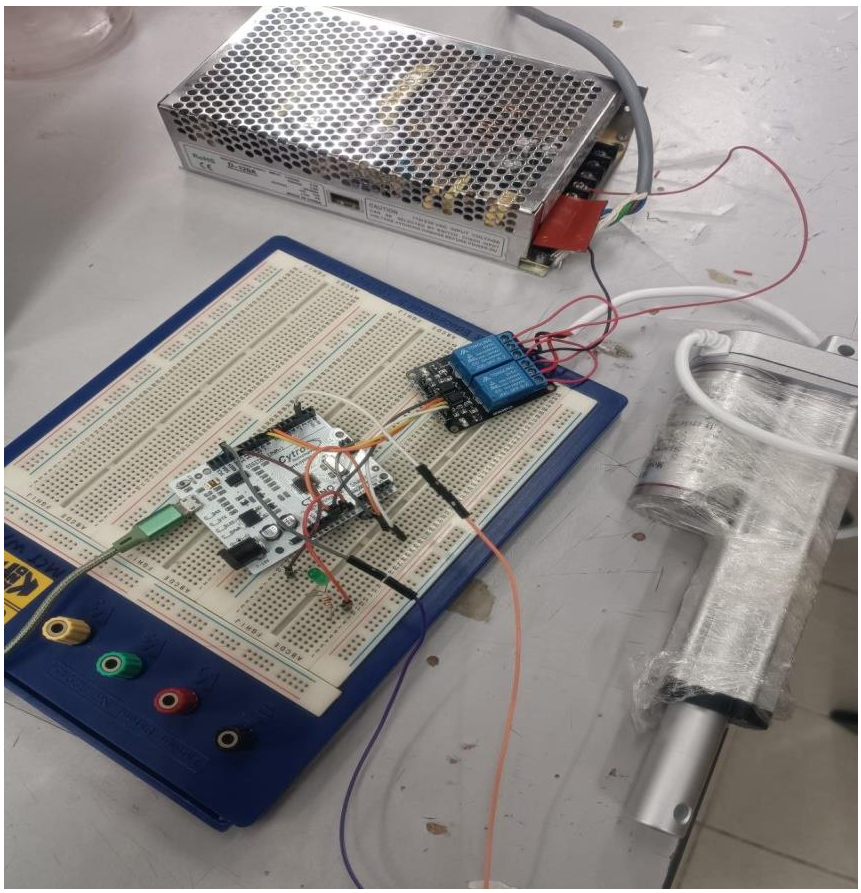

Fig.2 shows Connection of the 12V Linear actuator,LED,12V power supply, 2 way channel relay and Cytron CT-UNO

2.2 METHODOLOGY

a. Circuit assembly

- The positive of variable power supply is being connected to both NC pins of 2 way channel relay and the negative side is being connected to both NO pins of two way channel pins.
- The 12VDC linear actuator is being connected to COM pin on the relay.
- The VCC of 2 way channel relay is being connected to pin 5V and GND is connected to the GND of 2 way channel relay.
- The pin 8 of CYTRON-CT to pin IN2 and the pin 7 is being connected to pin IN1 of 2 way channel relay.

b. Programming and uploading

- The Arduino IDE software was opened on the computer.
- The correct board type (Arduino Uno) and communication port (COM port 9) were selected in the Tools menu.
- The coding was uploaded and compiled with no syntax errors

c. Programming logic

- When the actuator being retract the LED is turn off and when the actuator stops the LED is turn on.
- Both conditions applied when they hit the limit switch by using while =(digitalRead(limitswitch)==HIGH)
- When the actuator being extend the relay 1 is turn on and relay 2 is turn off by using digitalwrite(relay1,HIGH) digitalwrite(relay2,LOW)
- When the actuator being extract, the relay 1 is turn off and relay 2 is being turn on digitalwrite(relay1,LOW) digitalwrite(relay2,HIGH)
- When the actuator being stopped both relay 1 and relay 2 is being turn off.

d. Programming code

```
const int relay1 = 7;
const int relay2 = 8;
const int limitSwitch = 2;
const int led = 13;

void setup()
{
  pinMode(relay1, OUTPUT);
  pinMode(relay2, OUTPUT);

  pinMode(limitSwitch, INPUT_PULLUP);
  pinMode(led, OUTPUT);
}
void loop()
{
  extendActuator();
  while (digitalRead(limitSwitch) == HIGH)
  {
  }
  stopActuator();
  digitalWrite(led, HIGH); // Turn ON LED
  delay(5000);
  retractActuator();
  delay(5000);
  digitalWrite(led,LOW);
}

void extendActuator()
{
  digitalWrite(relay1,HIGH);
  digitalWrite(relay2,LOW);
}

void retractActuator()
{
  digitalWrite(relay1,LOW);
  digitalWrite(relay2,HIGH);
}

void stopActuator()
{
  digitalWrite(relay1, LOW);
  digitalWrite(relay2, LOW);
}
```

2.3 DATA COLLECTION

Limit Switch	Linear actuator	Led
The limit switch being hitted	The linear actuator going up	Turn on
The limit switch not being hitted	The linear actuator going down	Turn off

2.4 DATA ANALYSIS

The system used two relays (relay 1 and relay 2) to control the direction of the DC Motor inside the linear actuator. Using the extendactuator() function, we can see the actuator

will moving up when relay 1 (Pin 7) HIGH and relay 2 (Pin 8) LOW. Using the retractActuator() the relay 1 (Pin 7) LOW and relay 2 (HIGH), it will move the actuator going down .When the limit switch is hit, the code will goes to the while loop and will call stopActuator().The LED turns ON, and the system pauses for 5 seconds because delay (5000). When the LED turns off the loop () will starts over and extending the actuator again.

2.5 RESULT

Figure 1 shows the limit switch being hitted and the led is turn on

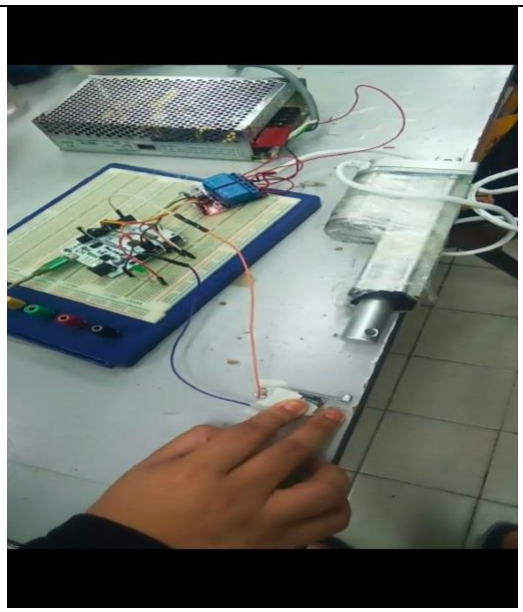

Figure 2 shows the limit switch not being hitted and the led turns off

2.6 DISCUSSION

The concept of H bridge is being used inside the codes to uses the two relays. This is because the DC motor needs to be moved in two directions and the polarity of the wires needed to be flipped. For example, during extension, when relay 1 is HIGH and the relay 2 is LOW, it will send the electricity in a single direction. The reason why the relays is being used rather than transistors, is because the high-voltage side is physically separated from the arduino's delicate logic side. Next, the relays are suitable because it doesn't get hot when passing the high current and making it great for movements that involved "Extend and Retract".

CONCLUSION

The first task successfully demonstrated that the fundamental principles of an H-bridge are essential for managing the bi-directional movement of a linear actuator. By transitioning from manual push-button switching to the L298P motor driver, the system gained the ability to handle higher current loads with greater reliability. Furthermore, the implementation of Pulse Width Modulation (PWM) proved to be an effective method for speed control; by adjusting the duty cycle of the signal, the actuator's extension and retraction speeds were finely tuned. This phase of the experiment confirmed that maintaining strict diagonal switching logic is vital to prevent electrical shorts and ensure the longevity of the motor driver.

In the second task, the integration of relays and a limit switch successfully transformed the open-loop system into a responsive, automated circuit. The use of Relay 1 and Relay 2 provided a robust electromechanical interface for reversing motor polarity, while the control logic ensured that the actuator could stop precisely upon reaching a physical limit. The inclusion of an LED indicator and a programmed delay provided necessary visual feedback and a safety buffer before the cycle restarted. Ultimately, this experiment validated that combining high-power switching components with sensor feedback creates a safe, repeatable, and autonomous motion control system suitable for practical engineering applications.

RECOMMENDATION

The first task successfully demonstrated that the fundamental principles of an H-bridge are essential for managing the bi-directional movement of a linear actuator. One of the recommendation is by transitioning from manual push-button control to an L298P motor driver, it enhances the system's ability to handle higher current loads with improved reliability, which is supported by studies using L298N drivers in PWM-controlled motor applications (YEŞİL & ŞAHİN, 2022). Furthermore, Pulse Width Modulation (PWM) is also widely recognized as an effective method for speed control by adjusting the duty cycle, allowing precise tuning of actuator extension and retraction speeds 75. Maintaining strict diagonal switching logic in the H-bridge is essential to prevent electrical shorts and protect

the motor driver, a principle emphasized in fault-tolerant H-bridge designs that ensure system reliability (Shengquan et al., 2022). According to YEŞİL & ŞAHİN (2022), PWM frequency and decay modes impact performance and stability, with certain frequencies optimizing flow rate stability in pump motors driven by L298N-based H-bridges. Overall, these findings confirm that using an L298P driver with PWM control and proper switching logic is a sound approach for reliable, efficient linear actuator operation.

For the second task, we can recommend that the system's reliability could be significantly improved by incorporating software debouncing for the limit switch and a fail-safe "timeout" mechanism in the code. In industrial environments, mechanical switches often produce "chatter" or false triggers due to vibration, which can lead to erratic behavior in the stopActuator() function. Furthermore, adding a secondary safety condition in the firmware that halts the motor if the limit switch is not triggered within a designated timeframe (e.g., 15 seconds) would prevent the motor from stalling and overheating in the event of a mechanical blockage or sensor failure.

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ACKNOWLEDGEMENT

We would like to sincerely thank the instructors, Assoc. Prof. Eur. Ing. Ir. Ts. Gs. Inv. Dr. Zulkifli Bin Zainal Abidin and the lab technicians, for all of their help, encouragement and support during this project. Their knowledge and perceptions have greatly influenced the course of this work. Additionally, we would like to express our gratitude to our peers for their support and cooperation, both of which were crucial to the accomplishment of this project.

STUDENTS DECLARATION

We hereby declare that, except for the places where it is acknowledged, all of the work presented in this report is entirely ours. We certify that, in completing this project, we have complied with the standards of academic integrity and have not engaged in any plagiarism or unethical behaviour. Every information and support source used in this work has been appropriately referenced and acknowledged.

Certificate of Originality and Authenticity

This is to certify that we are responsible for the work submitted in this report, that the original work is our own except as specified in the references and acknowledgement, and that the original work contained herein has not been done by unspecified sources or persons. We hereby certify that this report has not been done by only one individual, and all of us have contributed to the report. The length of contribution to the reports by each individual is noted within this certificate. We also hereby certify that we have read and understand the content of the total report and no further improvement on the reports is needed from any of the individual's contributor to the report. We therefore agreed unanimously that this report shall be submitted for marking, and this final report has been verified by us.

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